FINAL
PRESENTATION

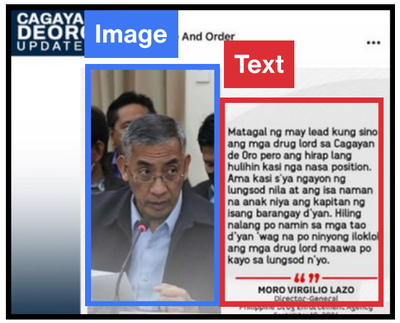
# EARLY DETECTION OF MULTIMODAL FAKE NEWS

## VIA REINFORCED PROPAGATION PATH GENERATION

L. ZHANG ET AL. · IEEE TKDE 37(2), 2025

# Introduction

Time




News
(Fabricated)

Share

Share

Share

User#01

User#07

User#25

Propagation path

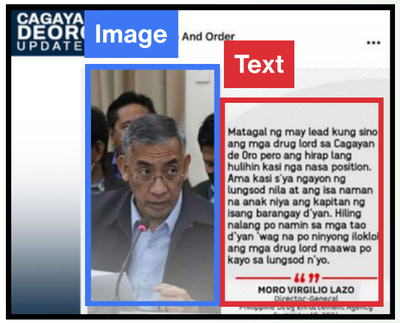
# Introduction

Time

## Content-based Detection

- Real Time
- Accuracy

Hazard Level: Low




News
(Fabricated)

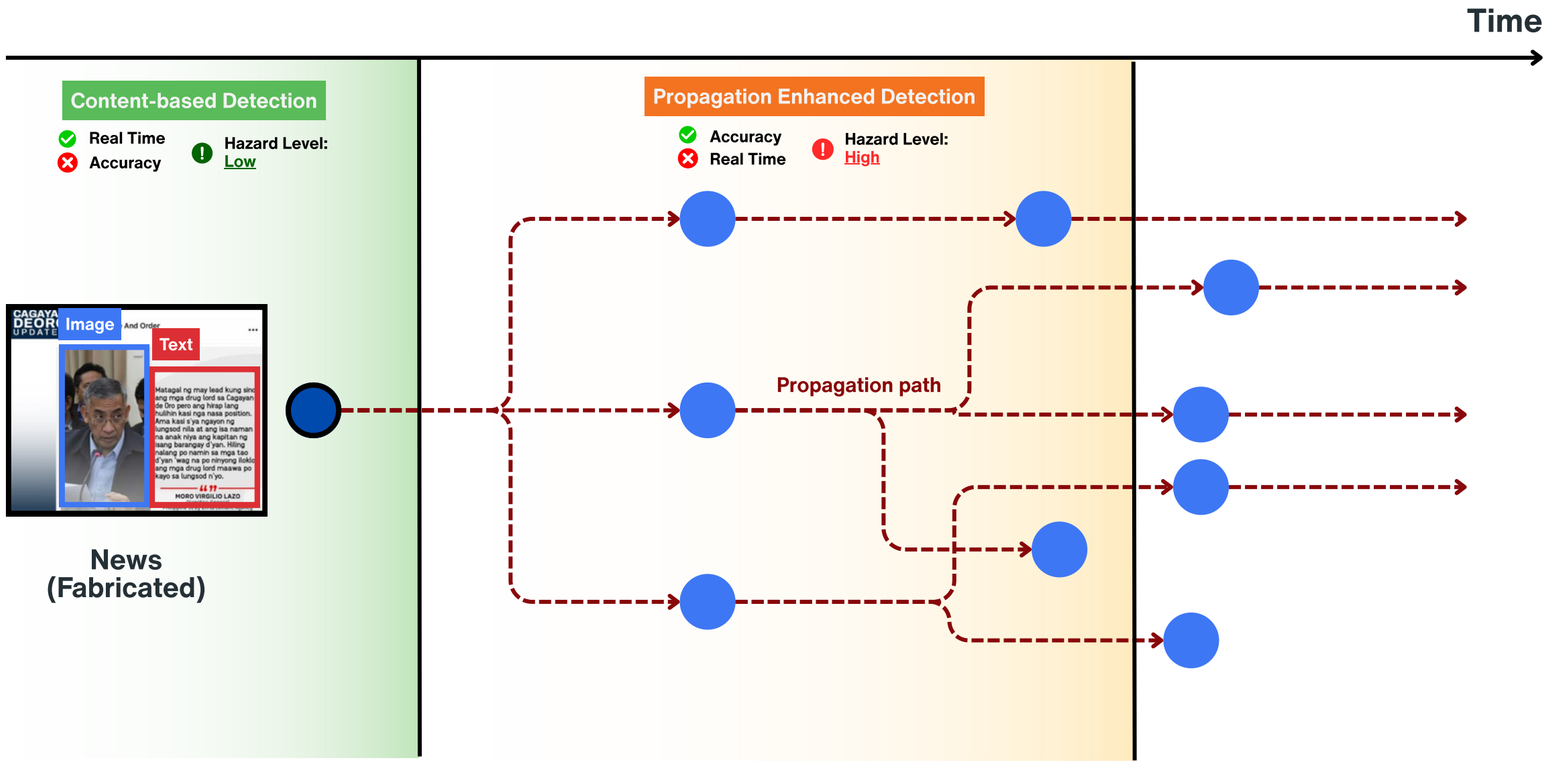
## Propagation Enhanced Detection

- Accuracy
- Real Time

Hazard Level: High

Propagation path

# Introduction




*Can we leverage the **abundant** social context in the **propagation paths** to enhance the **early-stage detection** before widespread dissemination ?*

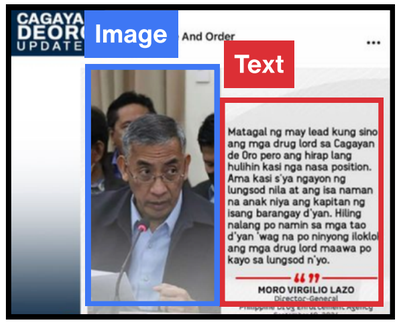
# Introduction

Time

**Reinforced Propagation Path Generation Fake News Detection (RPPG-Fake)**

- Accuracy
- Real Time
- Hazard Level: Low

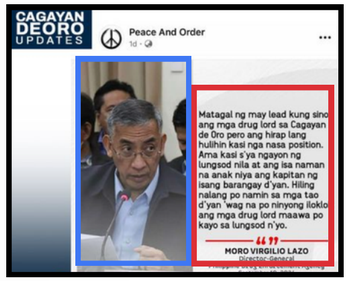
RL Generator




News (Fabricated)

Propagation path

Generate the propagation path (G')

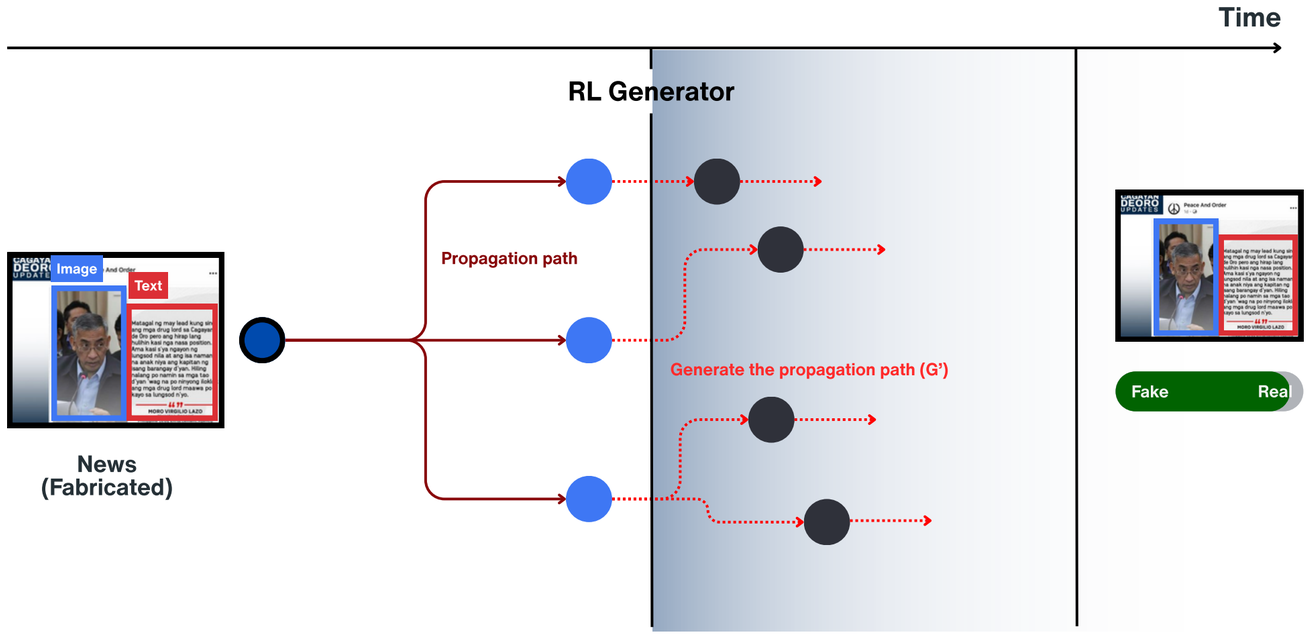



Fake Real

# Introduction

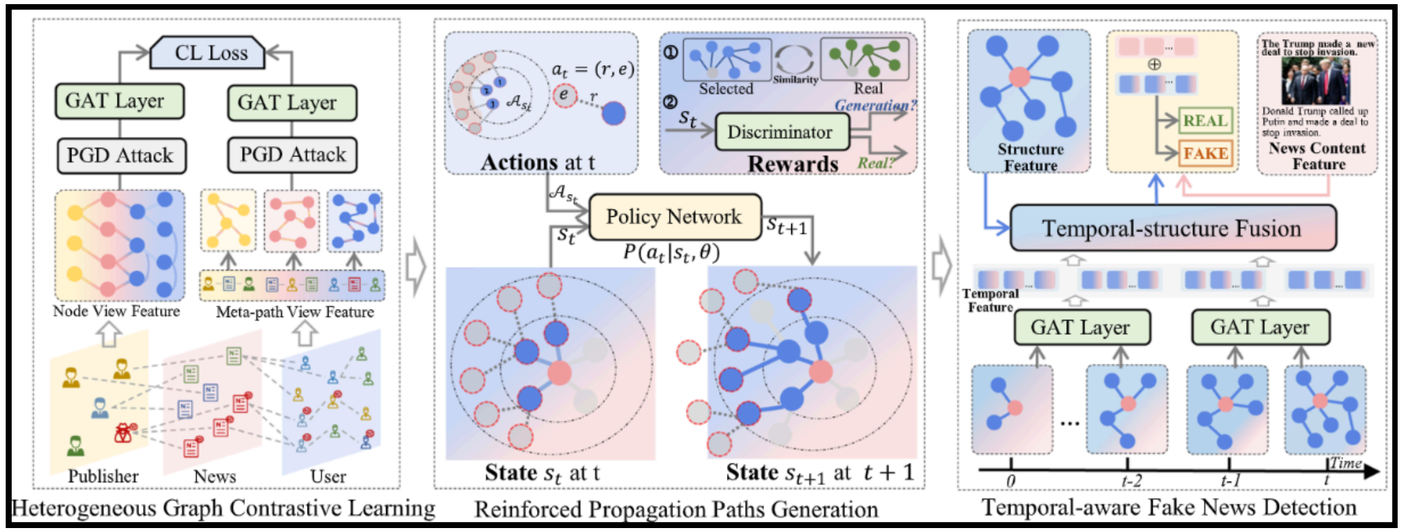
We are the first to explore the challenge of **early fake news detection by generation propagation paths**, which leverage the advantages of extensive social context while mitigating social hazards.

RPPG-Fake model has been proposed. The model can generate the reasonable propagation paths via **reinforcement learning, and furthur integrate propagation tropology structure and temporal depth information through a temporal-aware fake news detection module.**



# Methodology

$$P(H \mid P,N,U)\cdot P(G' \mid G,H)\cdot P(y \mid n,G') = P(y \mid n,G)$$



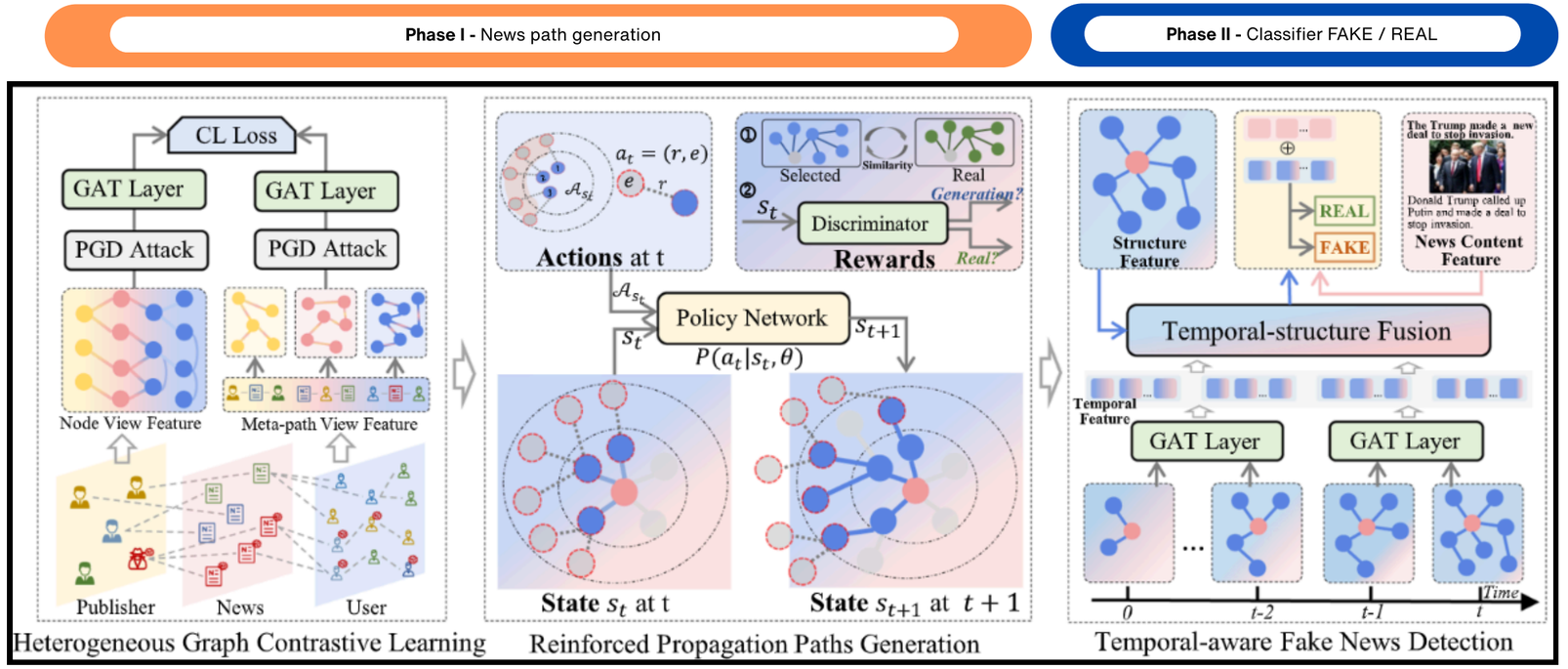

$$P(H \mid P,N,U)$$ → $$P(G' \mid G,H)$$ → $$\cdot P(y \mid n,G') =$$

**H:** Heterogeneous Graph
**P:** Publisher
**N:** News
**U:** Users

**G':** Generated Propagation Path
**G:** Early Propagation Path
**H:** Heterogeneous Graph

**y:** News Label ; 1 = Real, 0 = Fake
**n:** News details (Textual + Visual)
**G':** Generated Propagation Path

# Methodology



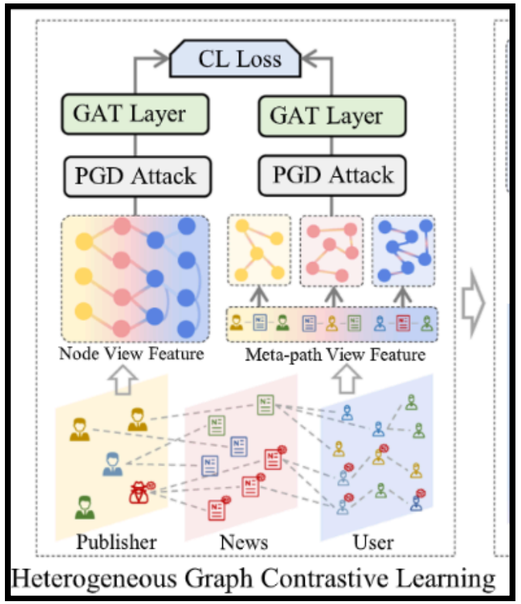

**Phase I -** News path generation

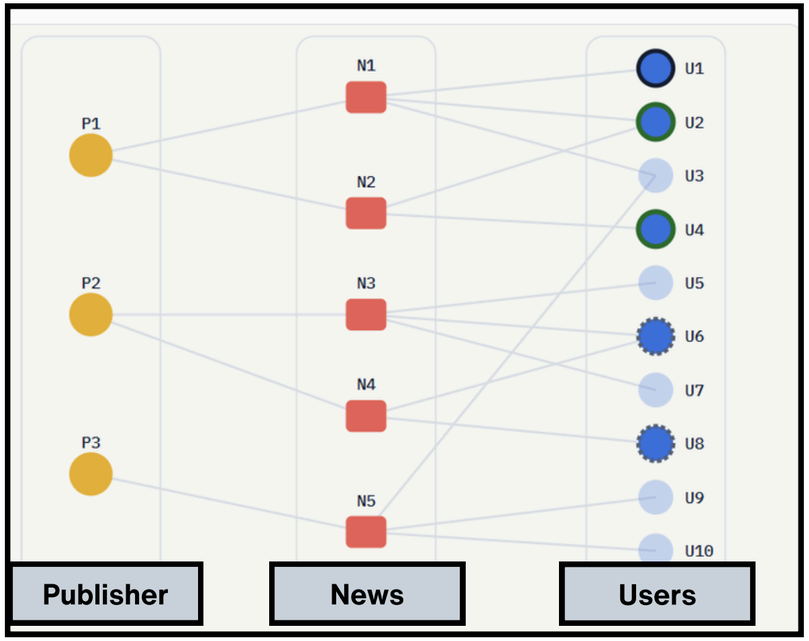
- Heterogeneous Graph Contrastive Learning
- Reinforced Propagation Paths Generations

**Phase II -** Classifier FAKE / REAL

- Temporal-aware Fake News Detection

# Methodology



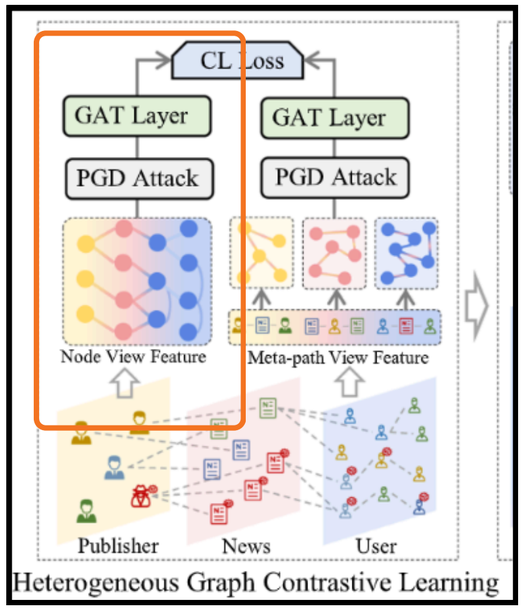

**Phase I -** News path generation - Heterogeneous Graph Contrastive Learning

**Construct the Heterogeneous Graph with 3 node types**—Publisher, News, and User—from platform-wide data, and **learn user embeddings without labels** through 2 views:

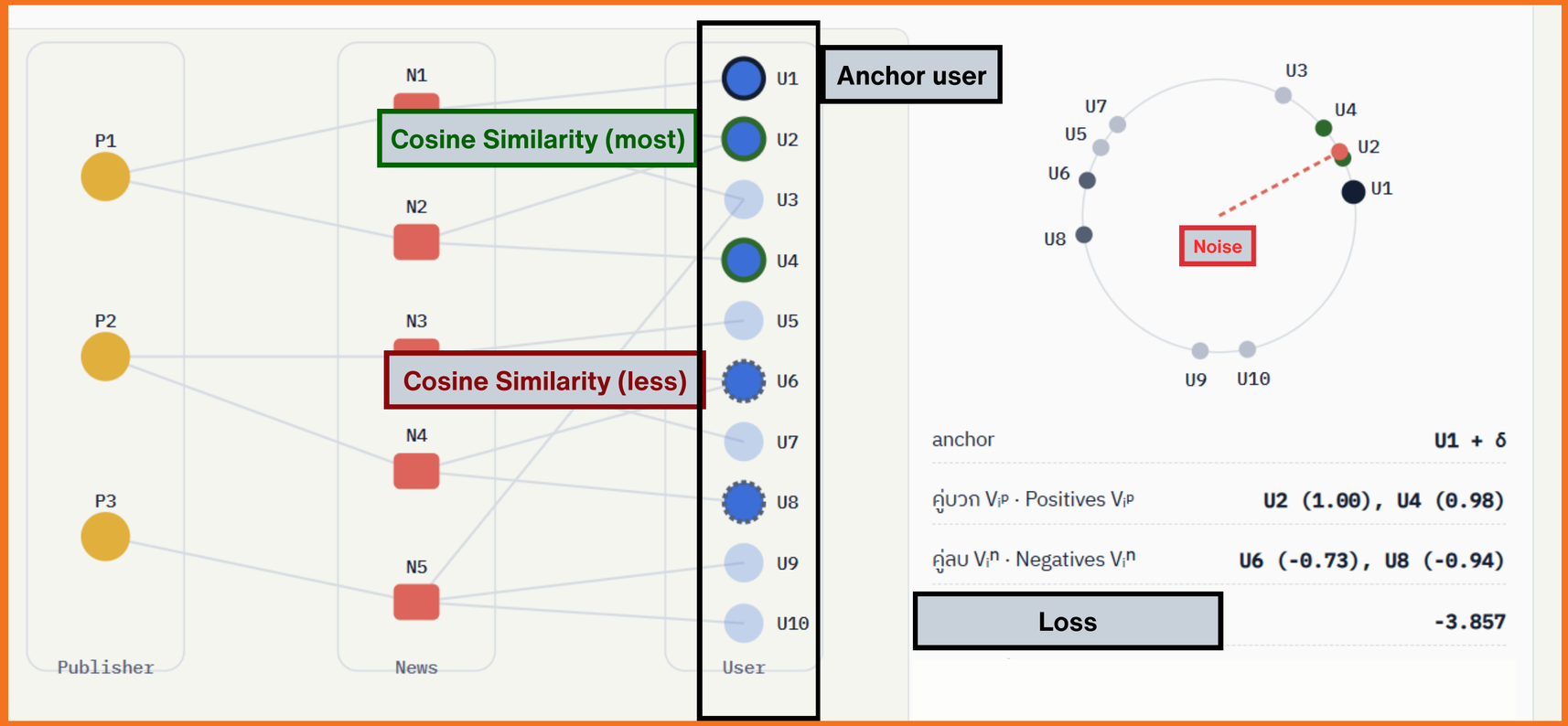
**Node View Contrastive Learning**

**Meta-path View Contrastive Learning**

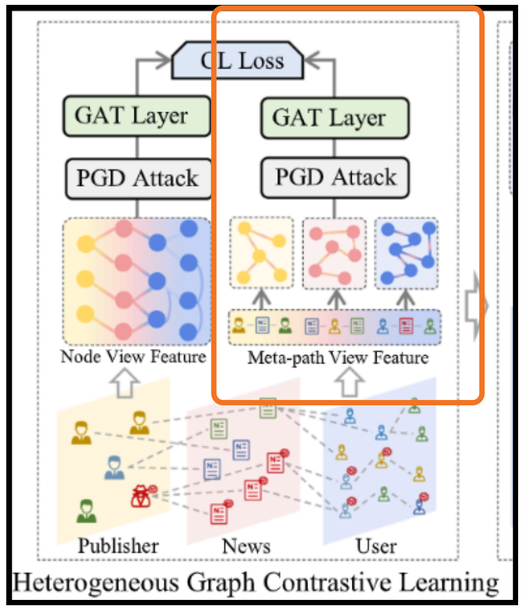
# Methodology



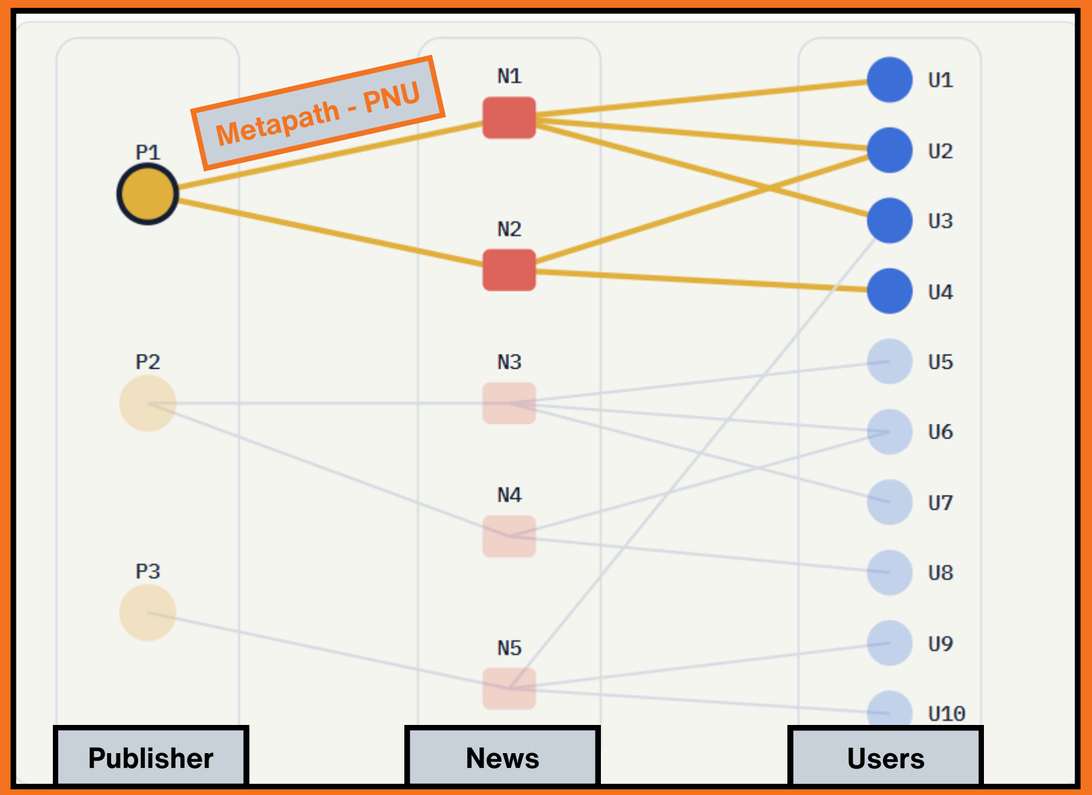

## Node View Contrastive Learning



# Methodology

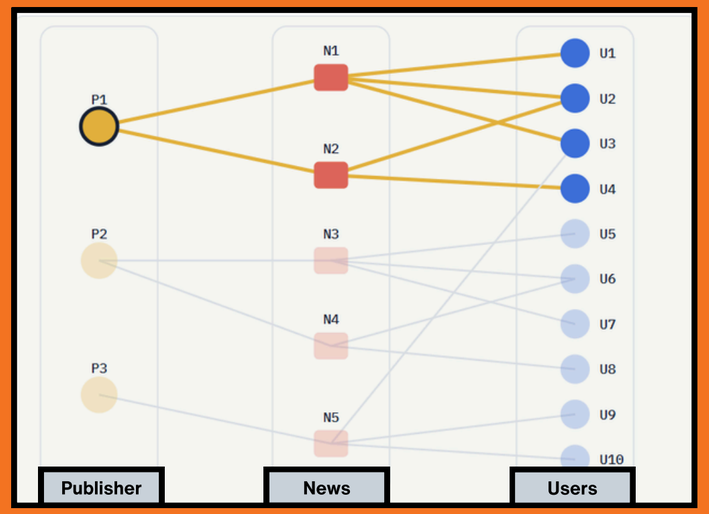



## Meta-path View Contrastive Learning

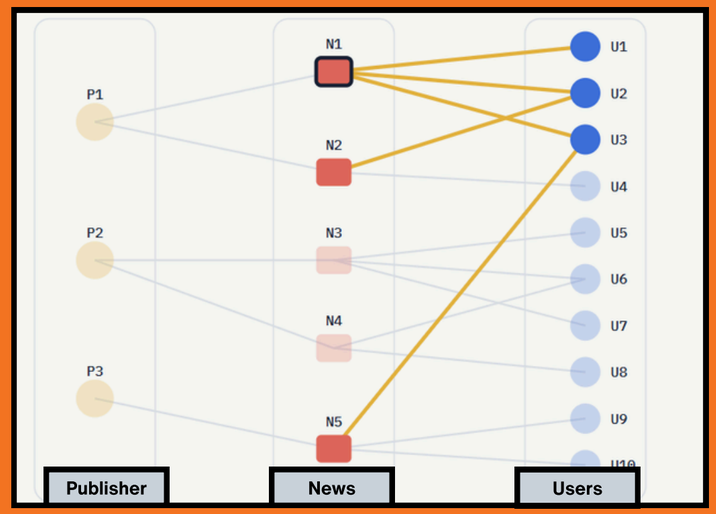


# Methodology

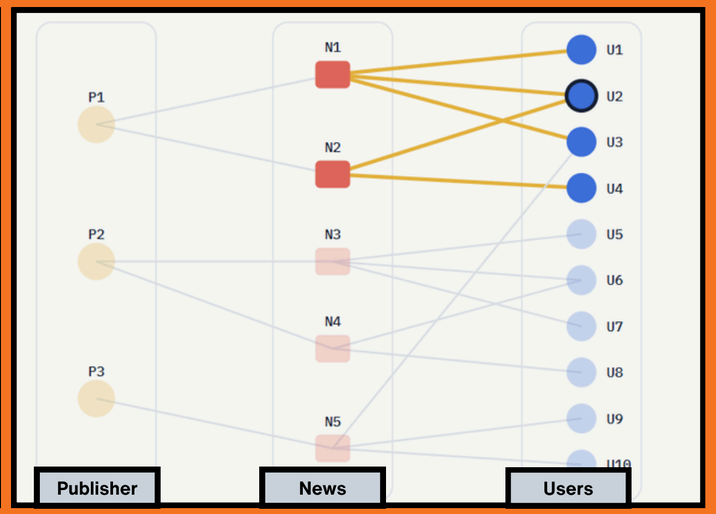
## Meta-path View ContrastiveLearning




P → N → U

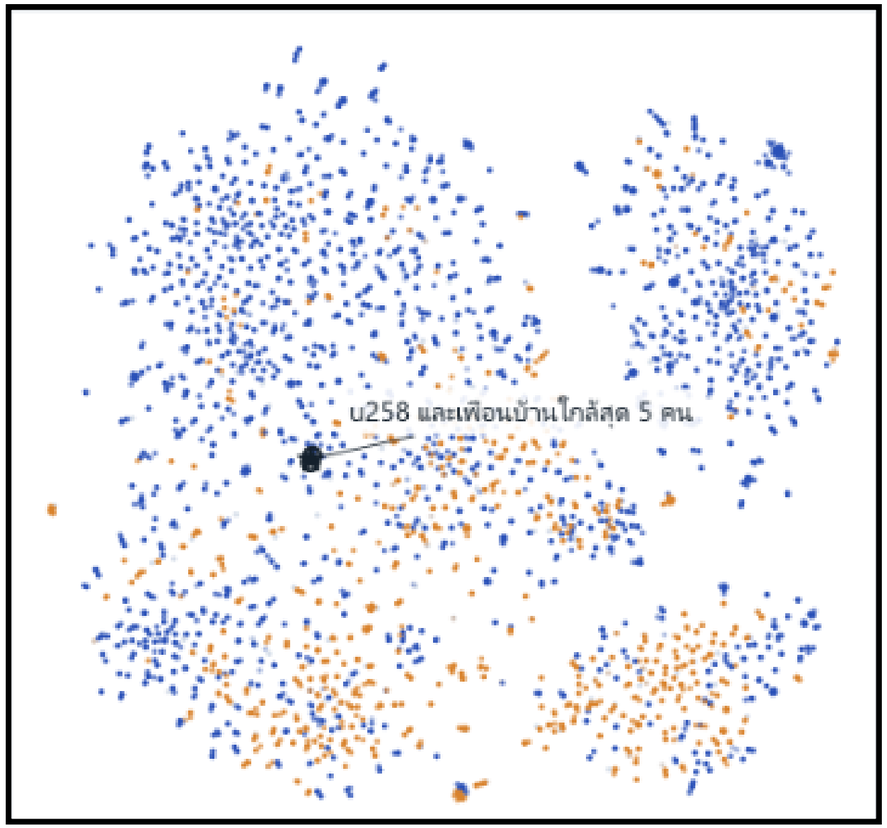


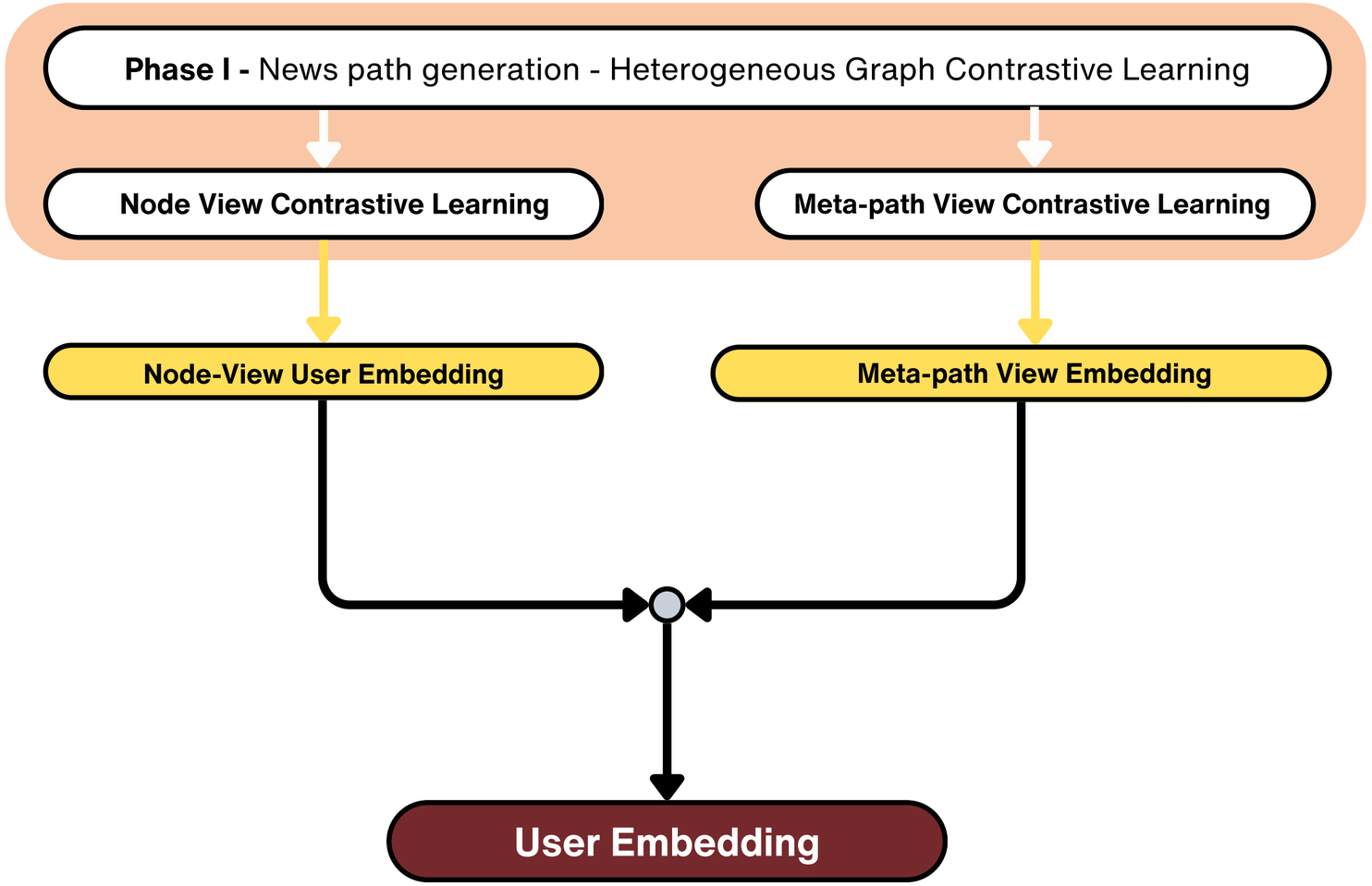

N → U → N



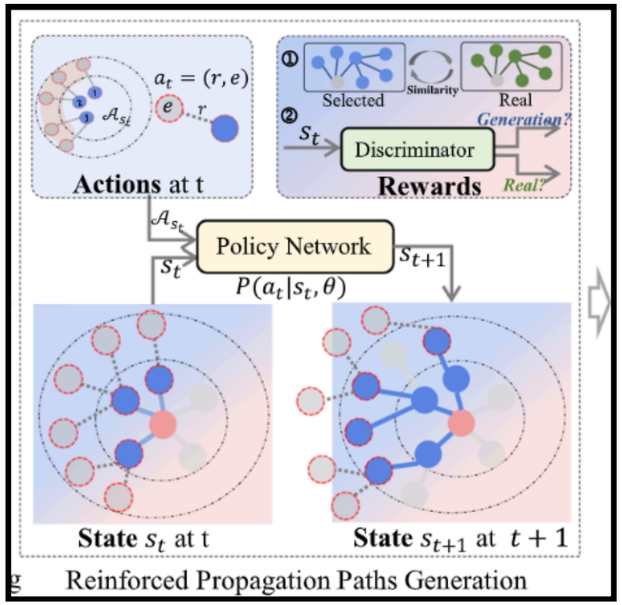

U → N → U

# Methodology



# Methodology




**Phase I -** News path generation - Reinforced Propagation Paths Generation

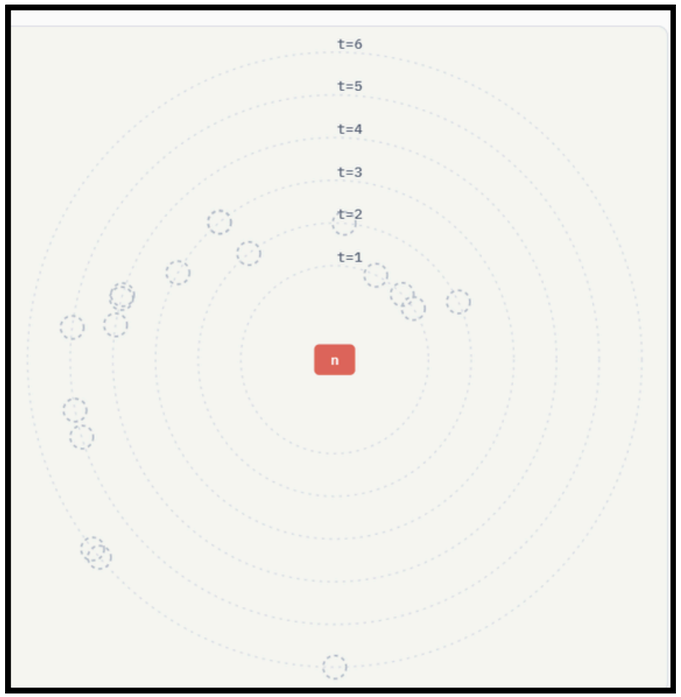
To simulate and generate **early-stage news propagation paths based on news content** without waiting for actual dissemination, thereby reducing social risks.

Policy Network

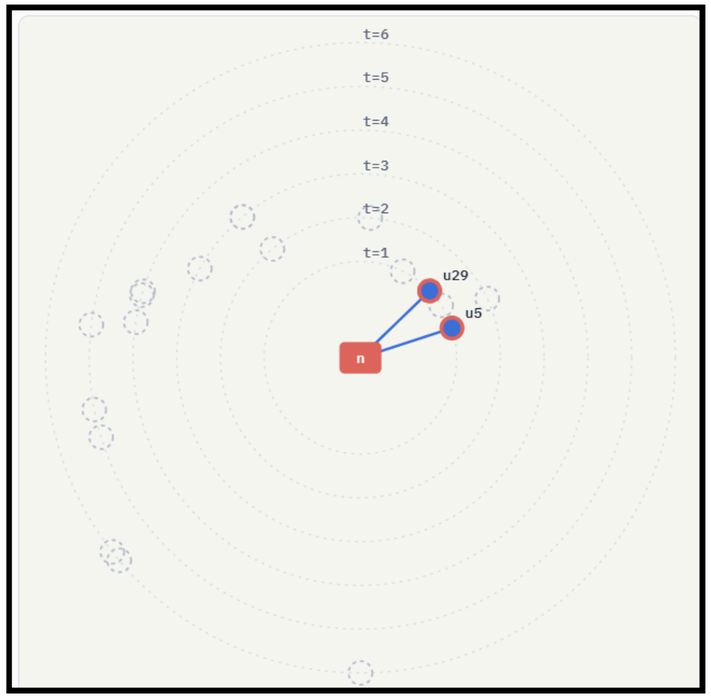
# Methodology

**Policy Network**

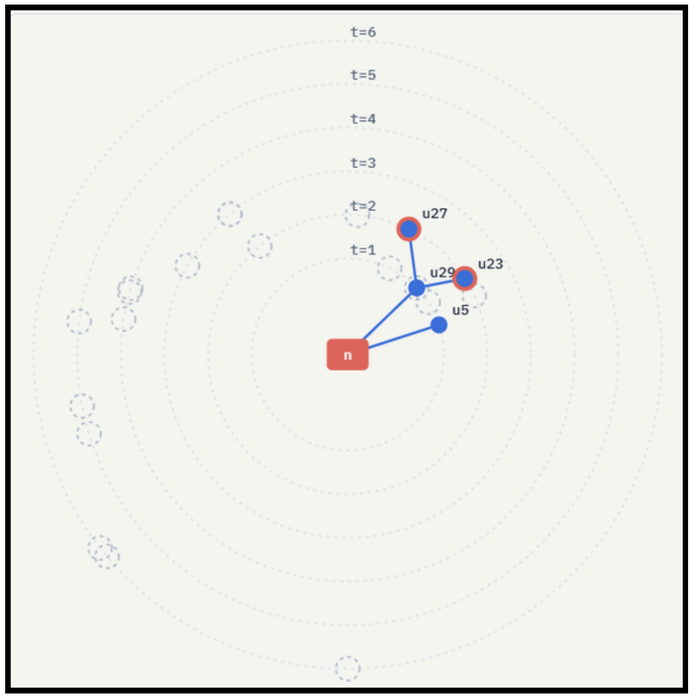
Time




t = 0



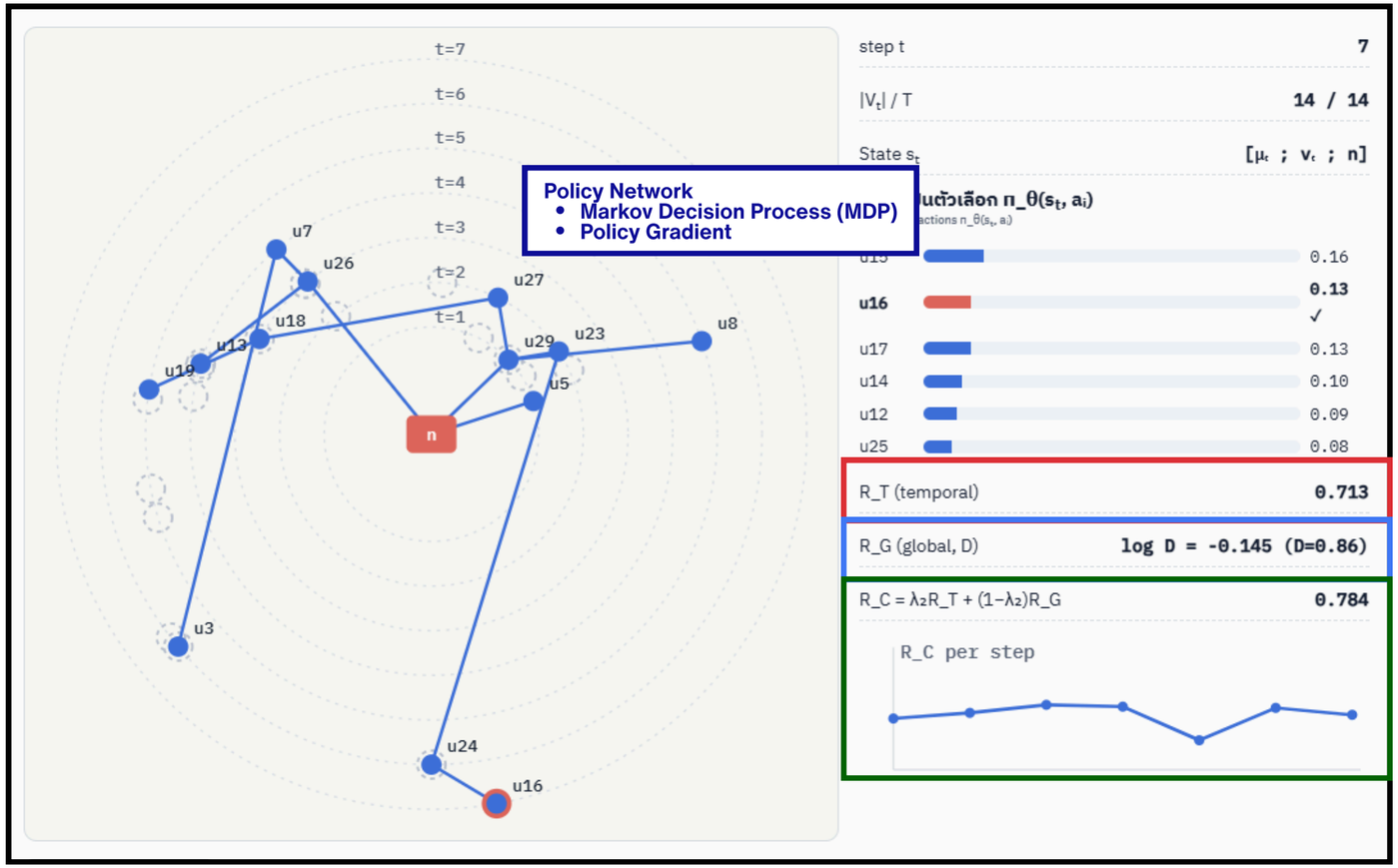

t = 1




t = 2

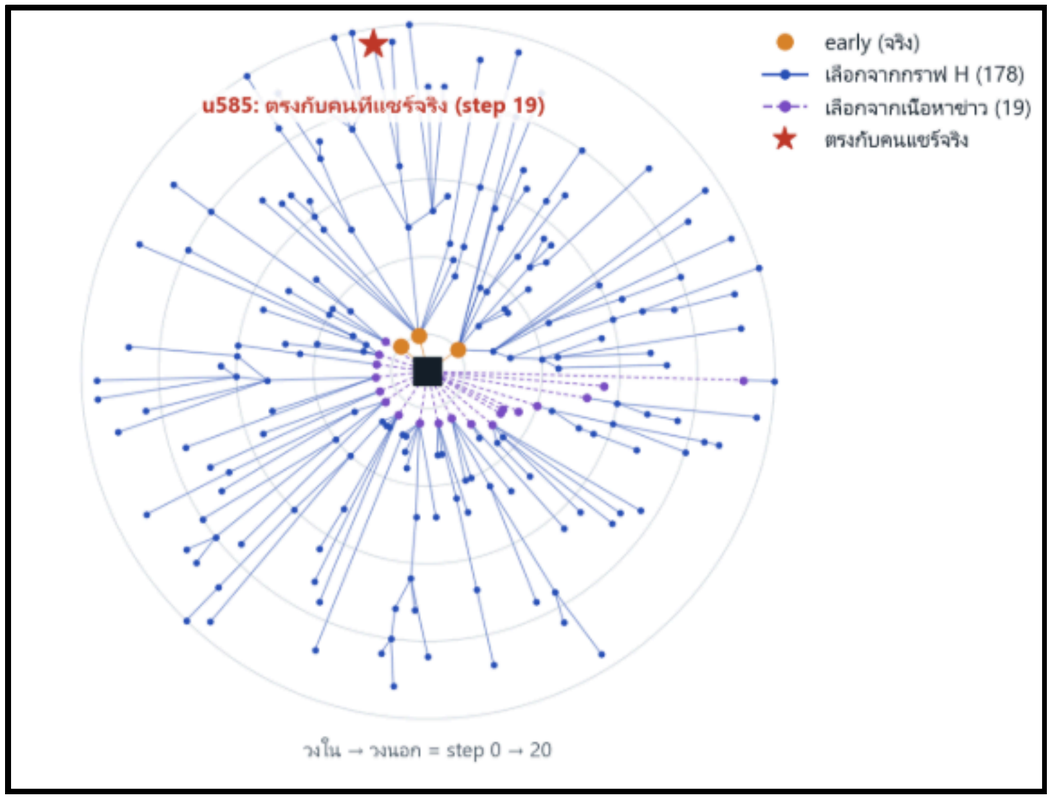
# Methodology

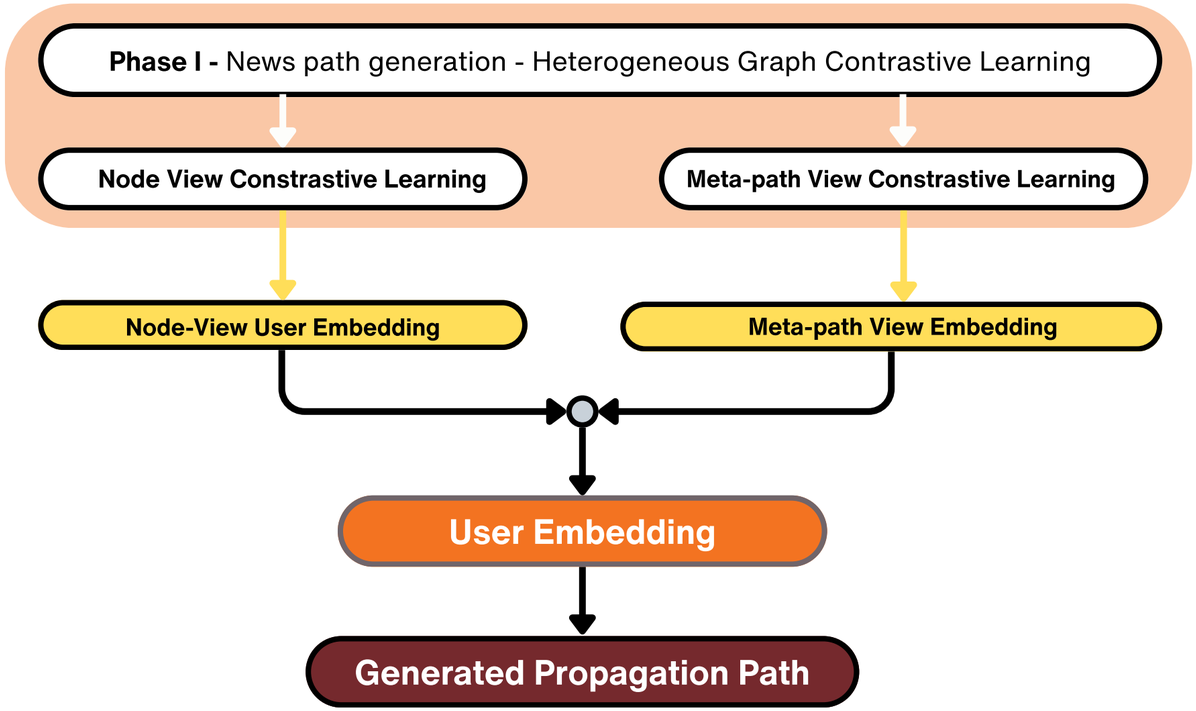
**Phase I -** News path generation - Reinforced Propagation Paths Generation



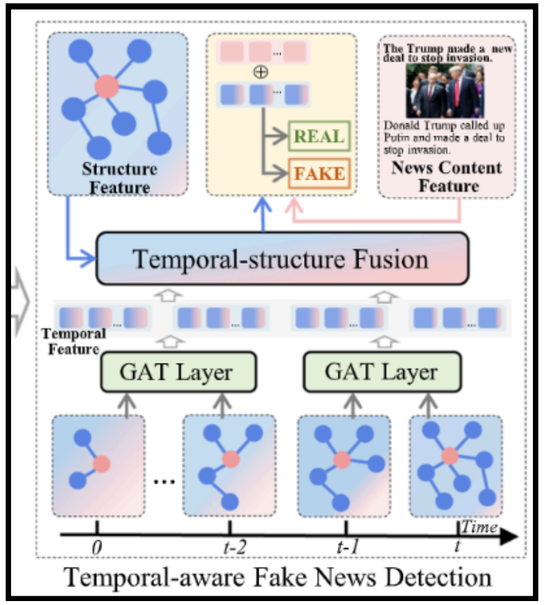
# Methodology

**Phase I -** News path generation - Reinforced Propagation Paths Generation



# Methodology




**Phase II -** Classifier - Temporal-aware Fake News Detection

Temporal-Aware Detection: **Dynamic Propagation Encoder**

Temporal-Aware Detection: Dynamic Propagation Encoder is a core sub-component of the RPPG-Fake model designed to **convert generated propagation paths into time-sensitive feature representations.**

Temporal-Aware Detection: **Fusion & Classification**

The final **classification component** of the RPPG-Fake framework.

# Methodology

Temporal-Aware Detection: **Dynamic Propagation Encoder**

**Self-Attention Mechanism**

**Early Stage**

Time

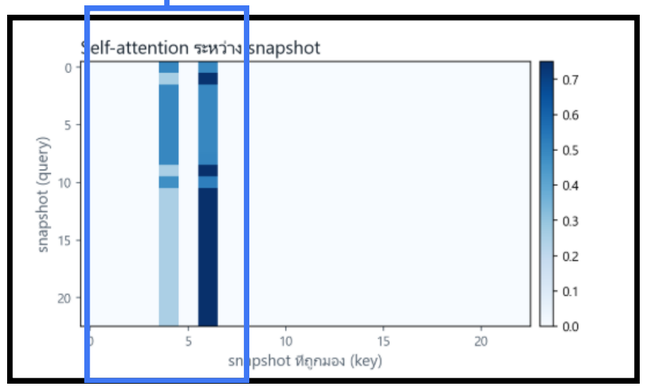
snapshot 0 · 2 node
early (จริง)

snapshot 2 · 4 node
early (จริง)

snapshot 7 · 54 node
+ G'

snapshot 14 · 124 node
+ G'

snapshot 22 · 201 node
+ G'




**Predicted Target (Key)** from the most attentions.

Generated User

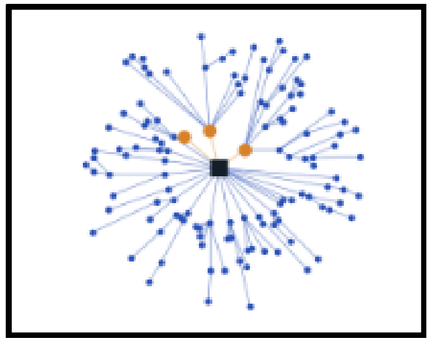
Real user

News

# Methodology

Temporal-Aware Detection: **Fusion & Classification**

## Fusion & Classification



**A single representative** vector of the overall propagation

+

**Visual content**

***BREAKING: Pope Francis Just Backed Trump, Released Incredible Statement Why- SPREAD THIS EVERYWHERE***

**Textual content**

*Trump is getting support from every leader, and that's the support that will make him grow great and strong. These elections will bring an immense change in our country.*

$$P(y \mid n, G)$$

**Approch to 1**: Predict as the **REAL** news
**Approch to 0**: Predict as the **FAKE** news

### Cross-Entropy Loss

This loss value will be used to update the weights of the entire model.

# Data

## Model training

**News**

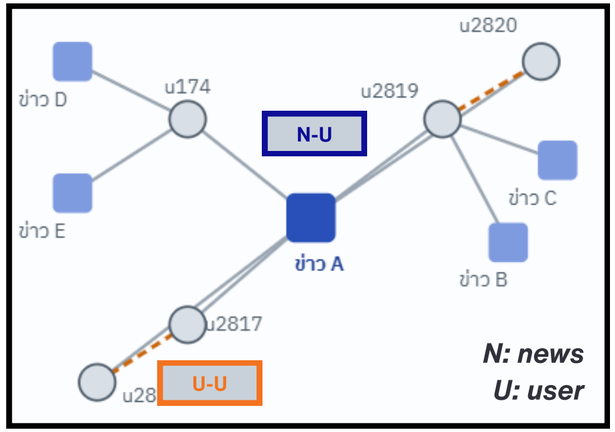
***BREAKING: Pope Francis Just Backed Trump, Released Incredible Statement Why- SPREAD THIS EVERYWHERE***

*Trump is getting support from every leader, and that's the support that will make him grow great and strong. These elections will bring an immense change in our country.*

**Visual content**

**Textual content**

**1 H graph / dataset**

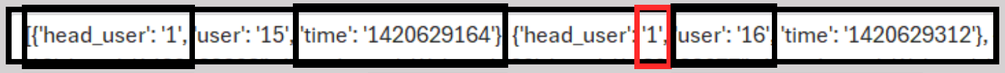



**Social Heterogeneous Graph**

**1 ID / dataset**

**Propagation path**

**User ID**




**Head user** | **Time** | **Destinated user**

# Data

Final Model

News

***BREAKING: Pope Francis Just Backed Trump, Released Incredible Statement Why- SPREAD THIS EVERYWHERE***

*Trump is getting support from every leader, and that's the support that will make him grow great and strong. These elections will bring an immense change in our country.*

**Visual content**

**Textual content**

**Early Propagation Path** *(3-5 shared)*

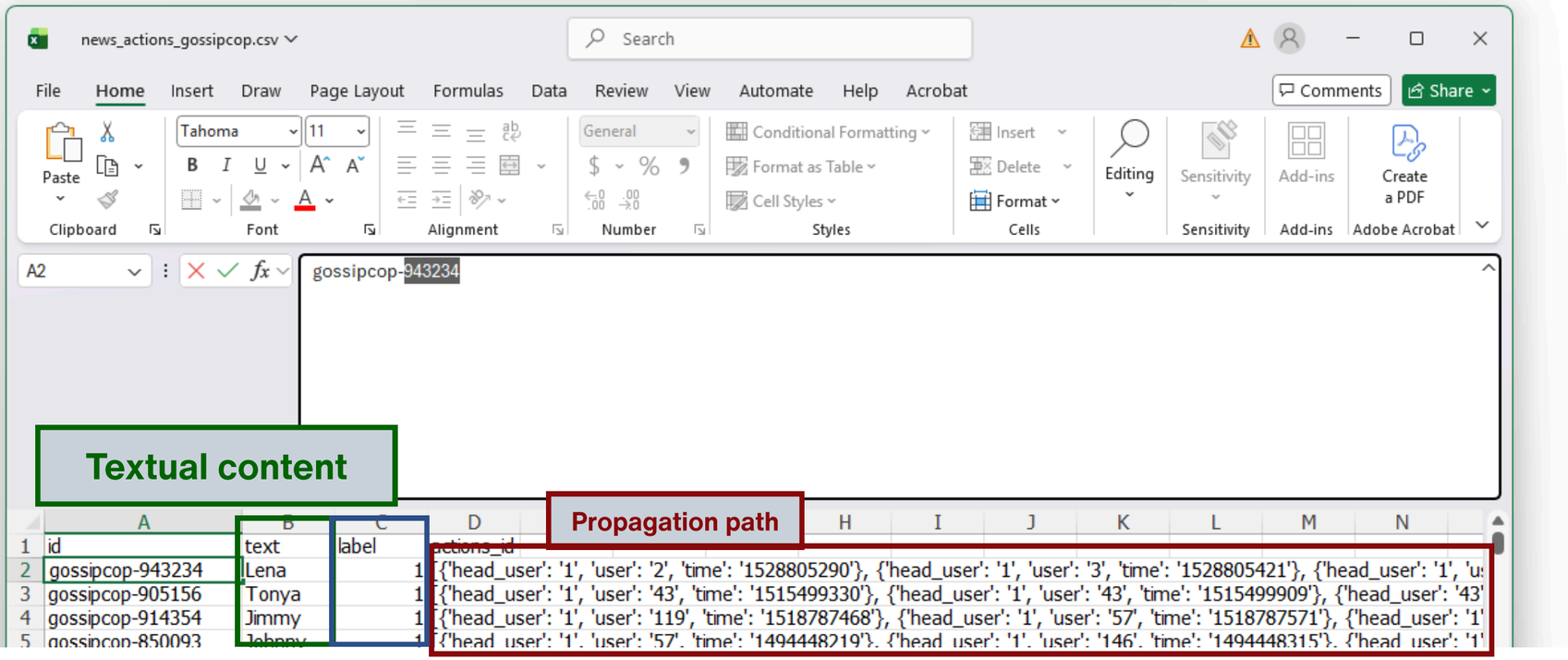
| index | id | label | text |
|---|---|---|---|
| 1937 | 5.44E+17 | fake | Aussie grand mufti & Imam Council condemn #Sydneysiege hostage capture http://t.co/m51P8dUPhB - LIVE UPDATES http://t.co/tTw3kMkYTZ |
| 1921 | 5.44E+17 | fake | Two female hostages have run out of #SydneySiege. 5 in total have escaped http://t.co/isvMtY4oUQ http://t.co/Xv2gJ1Q3zZ |
| 1991 | 5.45E+17 | fake | Hostages seen fleeing from cafe at centre of #SydneySiege moments ago http://t.co/nM2pss0I9d http://t.co/MXGOxDGdwv |
| 652 | 5.54E+17 | real | Zuckerberg on #CharlieHebdo: Facebook will protect free speech http://t.co/xHhVh8IXbX http://t.co/t10RSN5I2s |
| 204 | 5.53E+17 | real | A dark day for freedom of expression, an appalling human tragedy. #JeSuisCharlie #CharlieHebdo http://t.co/IaTCXg8iD4 |
| 690 | 5.54E+17 | real | Smoke, gunshots reported at site of standoff with suspects in the attack on Charlie Hebdo http://t.co/LDp3chB8FY http://t.co/IuAaEnrlXv |

# Input (Train)



gossipcop-943234.jpg

Visual content




**y:** News Label ; 1 = Real, 0 = Fake

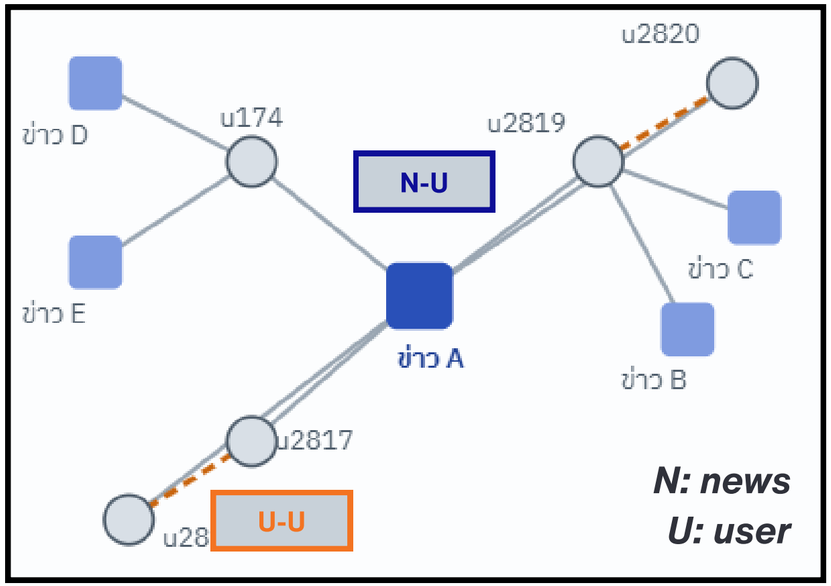
# Data

## Input (Train)

**build_graph(),** *a command.*

```
# src/data/components/politifact.py → build_graph()
NEWS_ROOT = 1                                   #
head_user = 1 หมายถึงตัวข่าวเอง
src, dst, rel = [], [], []
for i in train_idx:                             # ข่าว
train เท่านั้น
    news_node = num_users + i                   # node
ของข่าว = ต่อท้าย user
    users = set()
    for head, user, time in news[i]["eng"]:
        users.add(user)
        if head != NEWS_ROOT:                   # แชร์ต่อ
จาก user อื่น
            src.append(head); dst.append(user);
rel.append(0)   # 0 = repost (U-U)
    for user in users:
        src.append(news_node); dst.append(user);
rel.append(1)  # 1 = แชร์ข่าว (N-U)

edge_index = [src + dst, dst + src]             # เติม
เส้นย้อนทิศ
edge_type  = rel + rel
```




**Social Heterogeneous Graph**

# Experiment

| Dataset | Pheme | PolitiFact | GossipCop |
|---|---|---|---|
| #. total news | 3,506 | 478 | 5,526 |
| #. fake news | 1,142 | 294 | 1,935 |
| #. real news | 2,364 | 184 | 3,591 |
| #. words per news | 13.8 | 1545.8 | 578.6 |

| | |
|---|---|
| **Pheme** | This dataset is based from the **Twitter**, a widely used social media platform. |
| **Politifact** | This dataset is collected from **fact-check websites.** |
| **GossipCop** | This dataset is collected from **fact-check websites.** |

# Experiment

## Model baseline

### Content-based Detection Model

- **BERT** (2018)
- **MVAE** (2019)
- **EANN** (2018)
- **CAFE** (2022)
- **LLAMA2** (2022)
- **Detect-GPT** (2022)

### Propagation Enhanced Detection Model

- **CSI** (2017)
- **Bi-GCN** (2020)
- **UPFD** (2021)
- **MFAN** (2022)

# Experiment

## Implementation Details

All dataset are randomly split into training, testing, and validation set which following ratio.

**Holdout Split** (Random Split)

| **Train** (7) | **Test** (2) | **Val.** (1) |
| --- | --- | --- |
| For updating the **weight** and parameters in the model while training. | For testing the model performance using the unseen data | For tuning the performance while model is training for adjust the hyperparameters. |

# Experiment

## Implementation Details

All dataset are randomly split into training, testing, and validation set which following ratio.

**Holdout Split** (Random Split)

**Validation** (1)

- **AdamW** as the optimizer
  - The batch size: 16
  - The initial learning rate: $5e^{-3}$
  - The hidden size of user embedding: 128

Implement based on

**PyTorch 1.13.0** on
**Linux server** equiped with
**GeForce RTX3090 GPUs**

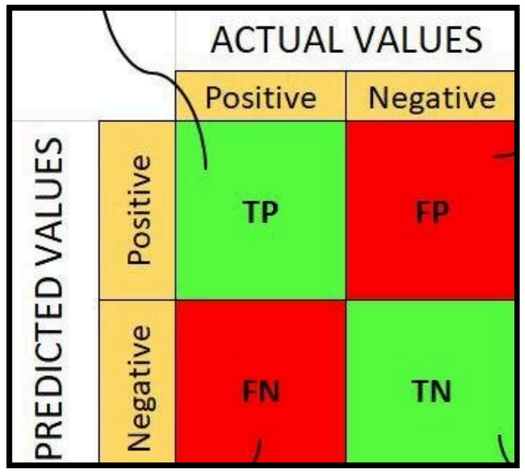
# Result

## Performance with **Full propagation path**

TABLE II
PERFORMANCE WITH FULL PROPAGATION PATHS

| Category | Method | PolitiFact | | | | GossipCop | | | | Pheme | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Acc | Pre | Rec | F1 | Acc | Pre | Rec | F1 | Acc | Pre | Rec | F1 |
| Propagation Enhanced | CSI | 0.753 | 0.834 | 0.742 | 0.758 | 0.782 | 0.823 | 0.783 | 0.808 | 0.771 | 0.825 | 0.831 | 0.828 |
| | Bi-GCN | 0.788 | 0.834 | 0.742 | 0.758 | 0.890 | **0.928** | 0.848 | 0.890 | 0.827 | 0.840 | 0.834 | 0.835 |
| | UPFD | 0.904 | 0.923 | 0.857 | 0.889 | 0.879 | 0.906 | 0.931 | 0.918 | – | – | – | – |
| | MFAN | 0.872 | 0.841 | 0.881 | 0.860 | 0.862 | 0.903 | 0.910 | 0.906 | 0.823 | 0.878 | 0.847 | 0.862 |
| **Proposed** | **RPPG-Fake** | **0.924** | **0.925** | **0.902** | **0.914** | **0.891** | 0.911 | **0.940** | **0.925** | **0.844** | **0.891** | **0.868** | **0.879** |

Best results are in bold and second best results are underlined.




**Accuracy**

measures the percentage of **correct predictions out of total predictions.**

$$\frac{TP + TN}{TP + TN + FP + FN}$$

**Recall**

measures the ability of a **classification model to find all actual positive instances in a dataset**

$$\frac{TP}{TP + FN}$$

**Precision**

measures the exactness of a machine learning **model's positive predictions**

$$\frac{TP}{TP + FP}$$

**F1 Score**

measures a classification model's accuracy by **combining precision and recall** into a single harmonic mean

$$2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

# Result

## Performance with **Fewer Propagation Paths at early stage**

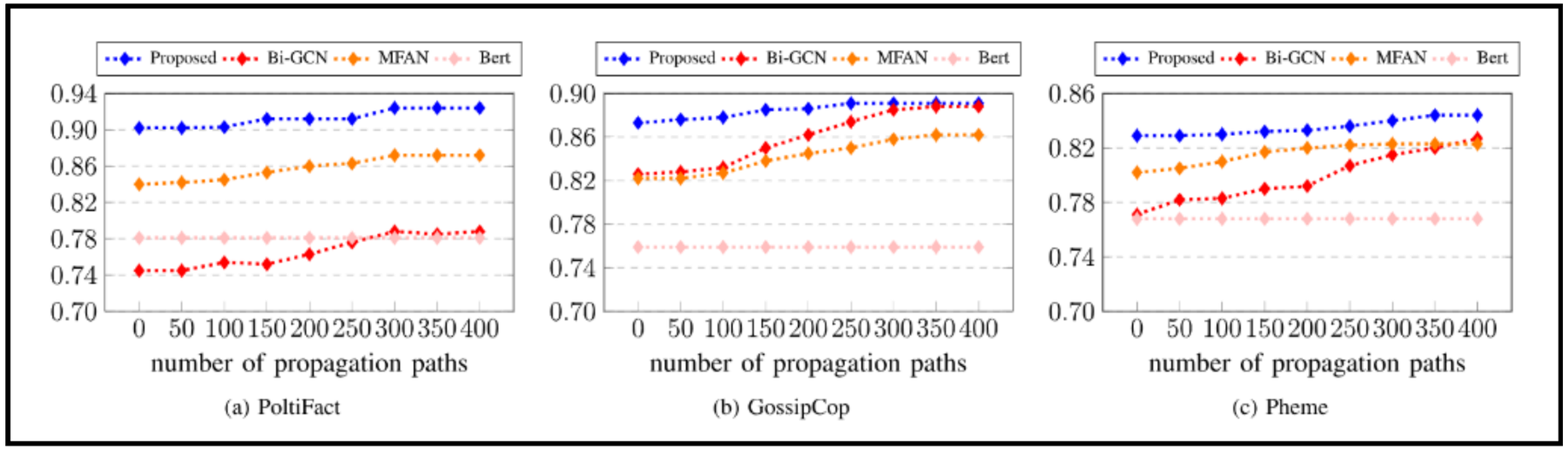
TABLE III
PERFORMANCE WITH FEWER PROPAGATION PATHS AT EARLY STAGE

| Category | Method | PolitiFact | | | | GossipCop | | | | Pheme | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Acc | Pre | Rec | F1 | Acc | Pre | Rec | F1 | Acc | Pre | Rec | F1 |
| Content Based | BERT | 0.781 | 0.766 | 0.878 | 0.818 | 0.759 | 0.787 | 0.764 | 0.775 | 0.768 | 0.704 | 0.778 | 0.739 |
| | MVAE | 0.812 | 0.803 | 0.835 | 0.819 | 0.782 | 0.802 | 0.751 | 0.776 | 0.776 | 0.734 | 0.722 | 0.728 |
| | EANN | 0.804 | 0.808 | 0.794 | 0.781 | 0.796 | 0.812 | 0.765 | 0.788 | 0.771 | 0.714 | 0.701 | 0.704 |
| | CAFE | 0.848 | 0.857 | 0.850 | 0.853 | 0.832 | 0.804 | 0.903 | 0.851 | – | – | – | – |
| | LLAMA2 | 0.848 | 0.865 | 0.780 | 0.820 | 0.825 | 0.811 | 0.872 | 0.840 | – | – | – | – |
| | Detect-GPT | 0.858 | 0.888 | 0.780 | 0.831 | 0.831 | 0.833 | 0.884 | 0.858 | – | – | – | – |
| Propagation Enhanced | CSI | 0.734 | 0.774 | 0.571 | 0.658 | 0.753 | 0.748 | 0.764 | 0.756 | 0.754 | 0.799 | 0.839 | 0.819 |
| | Bi-GCN | 0.745 | 0.696 | 0.762 | 0.727 | 0.826 | 0.871 | 0.881 | 0.876 | 0.771 | 0.825 | 0.831 | 0.828 |
| | UPFD | 0.862 | 0.892 | 0.786 | 0.835 | 0.822 | 0.860 | 0.891 | 0.875 | – | – | – | – |
| | MFAN | 0.840 | 0.846 | 0.787 | 0.815 | 0.822 | 0.861 | 0.888 | 0.875 | 0.802 | 0.853 | 0.846 | 0.850 |
| Proposed | **RPPG-Fake** | **0.902** | **0.921** | **0.853** | **0.886** | **0.873** | **0.904** | **0.921** | **0.913** | **0.829** | **0.881** | **0.854** | **0.867** |
| | *Imp.(%)* | *+4.0* | *+2.9* | *+0.3* | *+3.3* | *+4.1* | *+3.3* | *+1.8* | *+3.7* | *+2.7* | *+2.8* | *+0.9* | *+1.7* |

Best results are in bold and second best results are underlined.

# Result

## **The accuracy performance** with different generation users



# Ablation Study

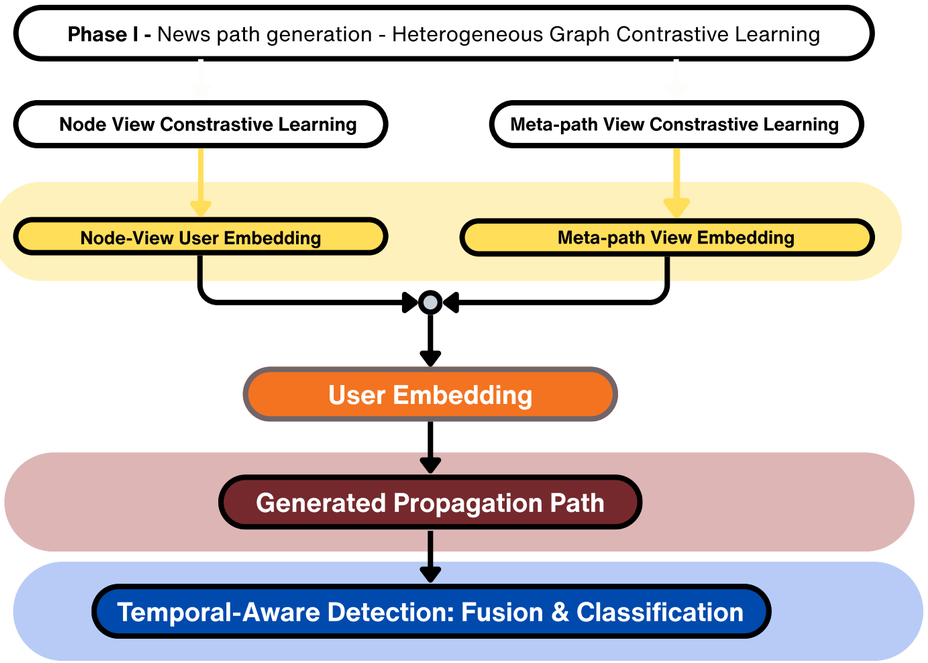



TABLE IV
ABLATION STUDIES OF *RPPG-FAKE* ON THREE DATASETS

| Abaltion Categoty | Method | PolitiFact | | | | GoosipCop | | | | Pheme | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Acc | Pre | Rec | F1 | Acc | Pre | Rec | F1 | Acc | Pre | Rec | F1 |
| **Proposed** | RPPG-Fake | **0.924** | **0.925** | **0.902** | **0.914** | **0.891** | **0.911** | **0.940** | **0.925** | **0.844** | **0.891** | **0.868** | **0.879** |
| **Heterogeneous Contrastive Learning** | w/o all CL | 0.891 | 0.919 | 0.829 | 0.872 | 0.874 | 0.910 | 0.921 | 0.915 | 0.827 | 0.879 | 0.852 | 0.867 |
| | w/o Node-view CL | 0.902 | 0.921 | 0.854 | 0.886 | 0.876 | 0.905 | 0.924 | 0.914 | 0.828 | 0.881 | 0.852 | 0.867 |
| | w/o Meta-path CL | 0.913 | 0.902 | 0.902 | 0.902 | 0.885 | 0.907 | 0.936 | 0.921 | 0.831 | 0.883 | 0.856 | 0.869 |
| **Propagation Path Generation** | Random Sample | 0.891 | 0.897 | 0.854 | 0.875 | 0.872 | 0.907 | 0.917 | 0.912 | 0.824 | 0.879 | 0.849 | 0.864 |
| | Node Similarity | 0.902 | 0.921 | 0.854 | 0.886 | 0.873 | 0.905 | 0.921 | 0.913 | 0.826 | 0.881 | 0.850 | 0.865 |
| **Temporal-aware Detection** | w/o structure feature | 0.913 | 0.923 | 0.878 | 0.900 | 0.886 | 0.908 | 0.937 | 0.922 | 0.834 | 0.885 | 0.858 | 0.871 |
| | w/o temporal feature | 0.902 | 0.900 | 0.878 | 0.889 | 0.884 | 0.906 | 0.935 | 0.920 | 0.836 | 0.887 | 0.859 | 0.873 |

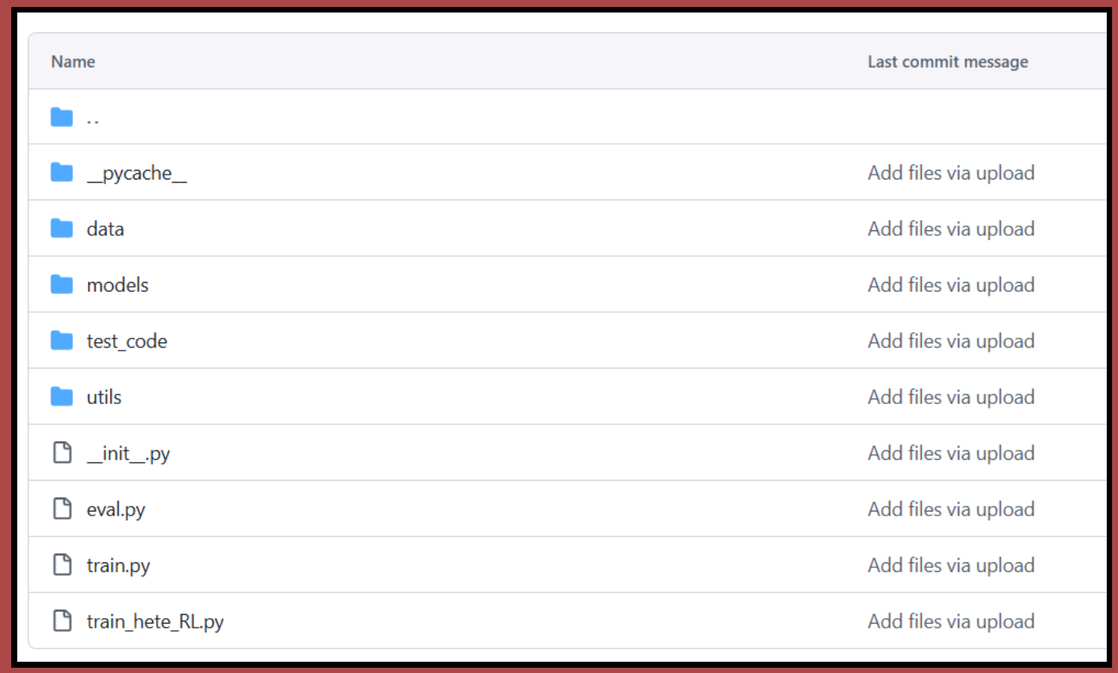
# Conclusion

- **Core Framework**
  - Introduces RPPG-Fake, a framework that simulates news propagation paths using **content features** and integrates them into a **temporal-aware detection module.**
- **Evaluation**
  - Validated across three popular benchmark datasets, demonstrating significantly improved detection performance.
- **Future Directions**
  - Suggests enhancing path generation with richer social dynamics, incorporating adaptive learning for better cross-domain generalization, and optimizing real-time performance and scalability for large-scale social networks.

# Limitation & Critique

## Supplementary Code

The Configuration files and some essential source code components from official GitHub are missing such as lacks code implementation for the Reward and Discriminator modules.

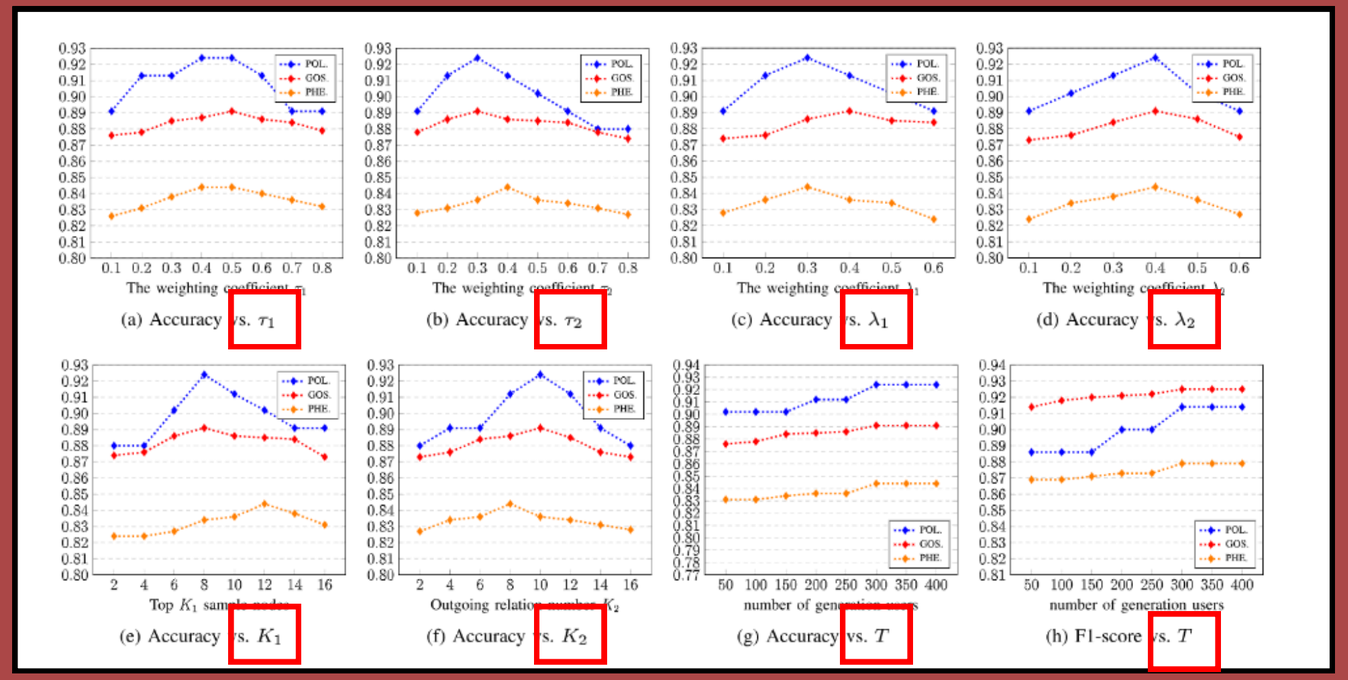


| Name | Last commit message |
|---|---|
| .. | |
| __pycache__ | Add files via upload |
| data | Add files via upload |
| models | Add files via upload |
| test_code | Add files via upload |
| utils | Add files via upload |
| __init__.py | Add files via upload |
| eval.py | Add files via upload |
| train.py | Add files via upload |
| train_hete_RL.py | Add files via upload |

# Limitation & Critique

## Parameter Setting

Some hyperparameter are only ploted without explicit numerical data. Including, key parameters are omitted, including RL epochs, dropout rates, GAT attention heads, and PGD values



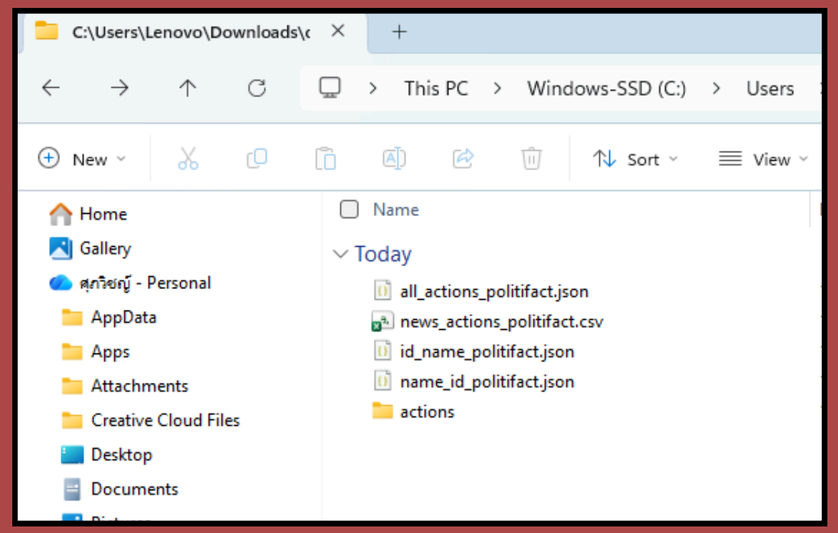

(a) Accuracy vs. $\tau_1$ (b) Accuracy vs. $\tau_2$ (c) Accuracy vs. $\lambda_1$ (d) Accuracy vs. $\lambda_2$

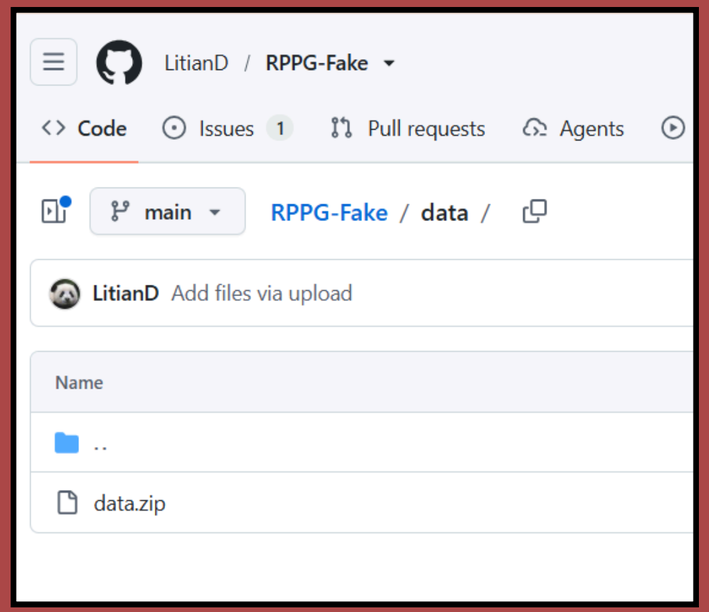
(e) Accuracy vs. $K_1$ (f) Accuracy vs. $K_2$ (g) Accuracy vs. $T$ (h) F1-score vs. $T$

# Limitation & Critique

## Incomplete Dataset

The provided repository contains only 483 news samples for PolitiFact. Including, lacks publisher information (all cascades share the same initial user node), making it impossible to construct Publisher-News-User (PNU) meta-paths. And GossipCop and PHEME datasets are not provided in the repository

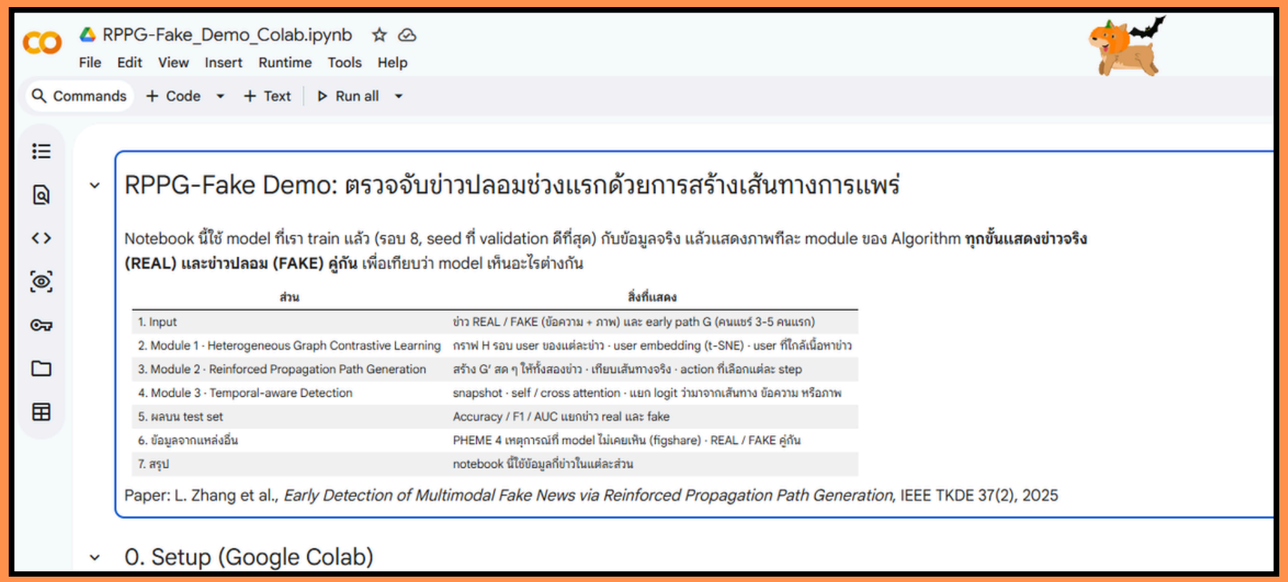


# Limitation & Critique

## Demo Limitation

From this demo about the code, the PC resource doesn't have much more. Therefore, some train runs will not based on the paper such as this demo using only 3 seed runs, 5 epochs.



# Limitation & Critique

| หัวข้อ | Paper | ของเรา (รอบ 8) |
|---|---|---|
| PHEME | 3,506 ข่าว | 2,004 ข่าว · เฉพาะที่มีครบ ข้อความ + ภาพ + เส้นทาง |
| GossipCop | 5,526 ข่าว · real 65% | 2,827 ข่าว · real 51% · ดึงเนื้อข่าวและภาพเอง |
| PolitiFact | 478 ข่าว · มีภาพ | 483 ข่าวของผู้เขียน · ไม่มีภาพ |
| Meta-path | PNU · NUN · UNU | UU · NUN · UNU (ข้อมูลไม่มี publisher) |
| Encoder | ไม่ได้เปิด code / weights | CLIP ViT-B/32 ทั้งข้อความและภาพ |
| จำนวนรอบ | 5 รอบ · รายงานค่าเฉลี่ย | 3 seeds · สุ่ม split ใหม่ทุก seed · รายงาน ± std |
| Module 1 / 2 | ไม่ระบุ step / epoch | contrastive 300 step · RL 5 epoch (รอบ 4 ใช้ 10) |
| Module 3 | ไม่ระบุ epoch | ≤ 30 epoch · early stop patience 10 · เลือกด้วย val acc |
| τ1 τ2 λ1 λ2 K1 K2 T | มีแค่กราฟ Fig. 4 | 0.4 · 0.3 · 0.4 · 0.4 · 8 · 10 · 200 (จุดสูงสุดของ Fig. 4) |
| แก้เพิ่ม | — | z-score ก่อน Eq. 7 · node view ใช้โครงสร้าง · user ใหม่ = UNK |
| เครื่อง | RTX 3090 · PyTorch 1.13 | RTX 5050 Laptop 8 GB · RAM 16 GB · PyTorch 2.11 |
| เหมือนกัน | AdamW lr 5e-3 · batch 16 · hidden 128 | split 7:1:2 · early stage = 3–5 engagements แรก |

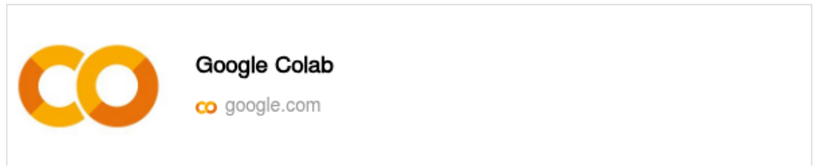
✕ Incomplete Dataset

✕ Parameter Setting

# Limitation & Critique

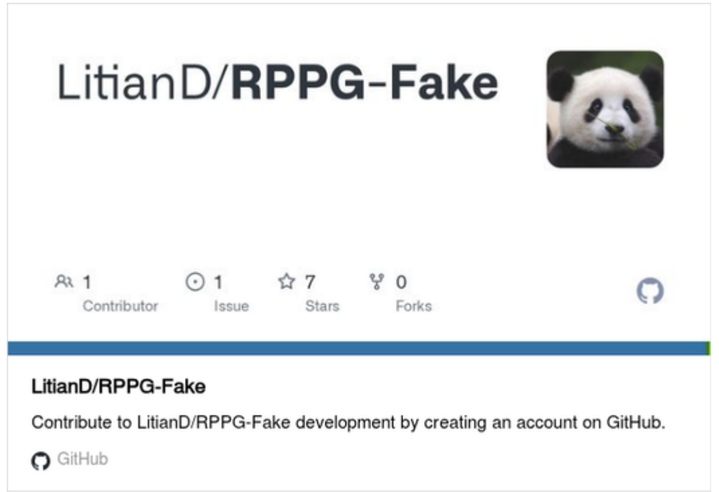
| Dataset | early Acc | paper | Δ pp | early F1 | paper | full Acc | paper | Δ pp | ไม่ใช้ G′ |
|---|---|---|---|---|---|---|---|---|---|
| PolitiFact | 0.826 ±0.04 | 0.902 | −7.6 | 0.781 | 0.886 | 0.868 | 0.924 | −5.6 | 0.868 |
| GossipCop | 0.878 ±0.09 | 0.873 | +0.5 | 0.879 | 0.913 | 0.908 | 0.891 | +1.7 | 0.900 |
| PHEME | 0.837 ±0.01 | 0.829 | +0.8 | 0.886 | 0.867 | 0.844 | 0.844 | 0.0 | 0.844 |

# Source Code




Google Colab

GitHub




Paper’s GitHub

THANK YOU

FINAL
PRESENTATION

# EARLY DETECTION OF MULTIMODAL FAKE NEWS

## VIA REINFORCED PROPAGATION PATH GENERATION

L. ZHANG ET AL. · IEEE TKDE 37(2), 2025

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